

AETHERIX

Autonomous Extraterrestrial High-throughput Enhancing Routing
and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN,
Quantum Communication, and Space-Based Infrastructure for Mars Mission Support

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EduQual Level 6 Diploma in AI Operations | Topic 59 | January 2026

matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

PRESENTATION AGENDA

01 The Challenge

Why interplanetary communication is hard

02 AETHERIX Architecture

DTN + AI Routing + Quantum Security

03 DTN & Bundle Protocol v7

Store-and-forward networking foundation

04 5-Tier Network Topology

241 nodes across Earth and Mars

05 Optical Link Budget

1550 nm laser performance analysis

06 RL-Based Routing

Multi-agent federated Q-learning

07 Quantum Security (QKD)

BB84/E91 with repeater chains

08 Orbital Mechanics

Contact windows and synodic period

09 Mars Mission Scenario

End-to-end simulation walkthrough

10 Performance Comparison

AETHERIX vs current systems

11 Roadmap & Standards

CCSDS, IETF, deployment phases

12 Conclusion & Q&A

Summary and live demo

WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

The Problem

- Earth-Mars: 54.6M – 401M km distance
- One-way light delay: 3 – 22 minutes
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work
- Current systems: 0.5 – 6 Mbps only

AETHERIX delivers a complete interplanetary networking solution

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

10-100×

Faster

>95%

Availability

241

Nodes

149

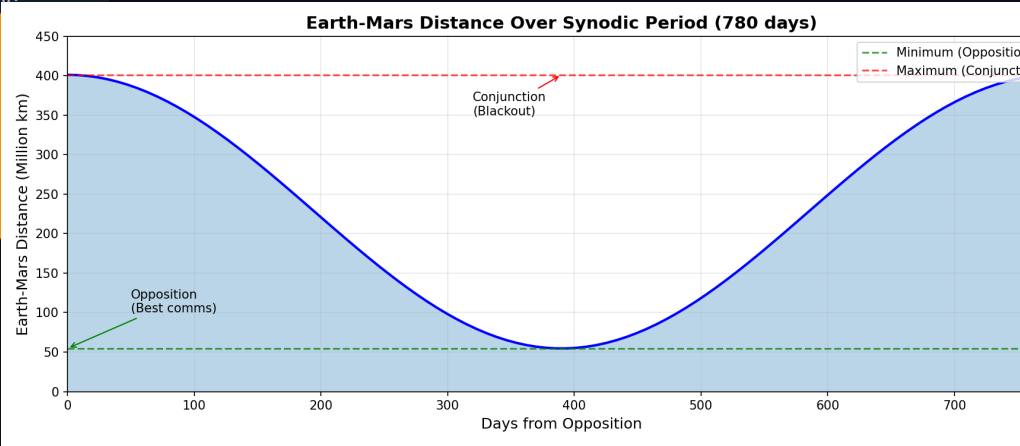
Tests

THE DISTANCE

Why Space Communication is Fundamentally Different

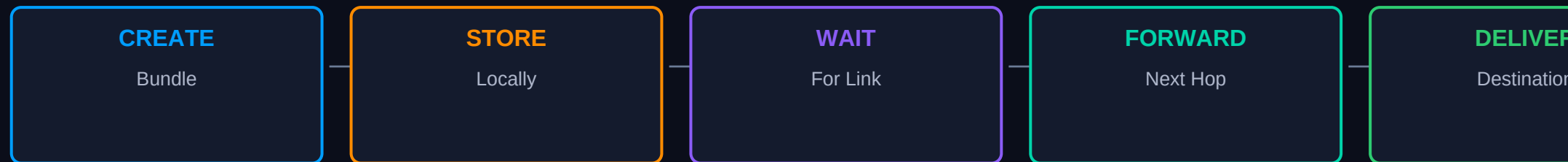
Earth → Mars	Distance	Light Time	Data Rate
Perihelion (closest)	54.6M km	~3 min	100-200 Mbps
Average	225M km	~12.5 min	10-20 Mbps
Aphelion (farthest)	401M km	~22 min	2-5 Mbps

TCP/IP Assumption	Space Reality	Impact
RTT < 1 second	6 – 44 min RTT	Timeout failure
Always connected	Scheduled contacts	Connection drops
End-to-end path	No persistent path	Routing impossible
Fast acknowledgments	ACKs take minutes	Window collapse
Low packet loss	High loss (optical)	Retransmit storms



THE ANSWER: DELAY-TOLERANT NETWORKING

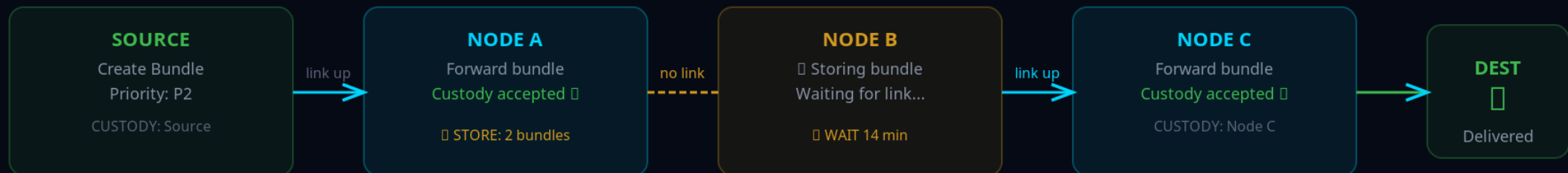
Decouple Communication from Connectivity



TCP/IP: END-TO-END CONNECTION (FAILS IN SPACE)



DTN: STORE-AND-FORWARD (WORKS IN SPACE)



Key: Custody Transfer

Each node takes responsibility for the bundle

Data Never Lost

Bundle stored until next hop confirms

No End-to-End Needed

Hop-by-hop, works with 22-min delays

Three Convergence Layers

LTP (RFC 5326) — Deep space

TCPCL (RFC 7242) — Earth segment

UDP-CL — Optical ISL

SYSTEM ARCHITECTURE

5 Integrated Modules for Interplanetary Communication

Infrastructure

Link budget calculations, optical/RF analysis

Routing

RL agents, BPv7 bundles, forwarding engine

Security

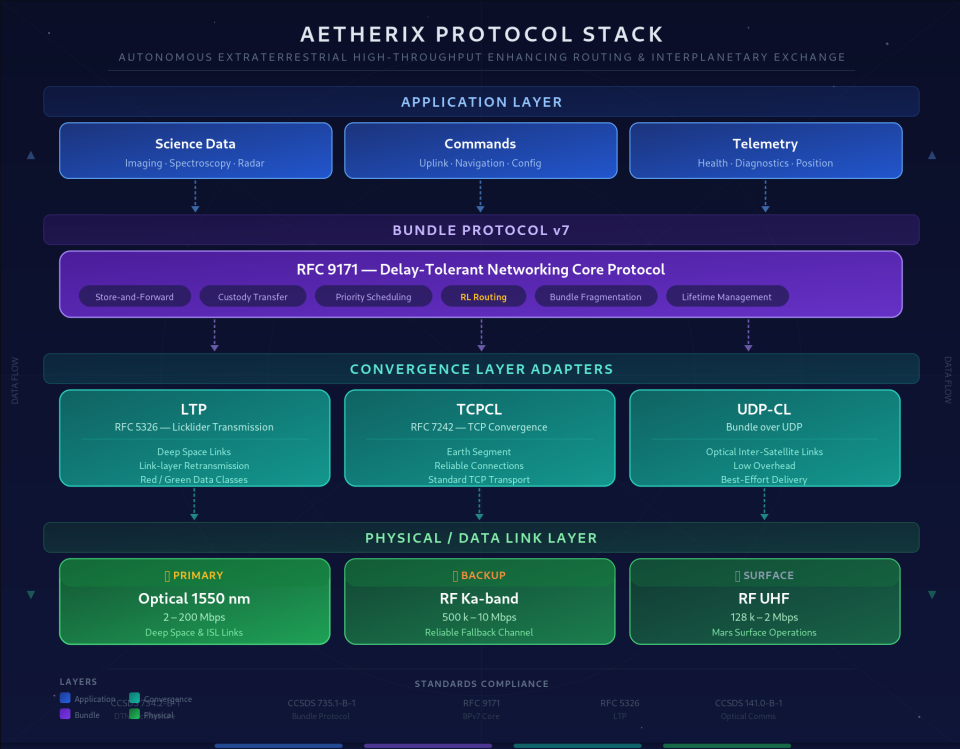
BB84/E91 QKD, repeater chains, privacy amplification

Orbital

Contact windows, Doppler, celestial body database

Simulation

End-to-end scenario engine, policy routing



Layer	Module	Engine	Showcase
Infrastructure	link_budget.py	OpticalLinkBudget	Earth-Mars scenarios
Routing	rl_agent.py	RLRoutingAgent	Federated Q-learning
Security	qkd.py	BB84/E91	Quantum key exchange
Orbital	contact_windows.py	ContactPredictor	Synodic period
Simulation	simulator.py	SimEngine	Full mission sim

SYSTEM ARCHITECTURE DIAGRAM

AETHERIX PLATFORM

INFRASTRUCTURE

link_budget.py
rf_link_budget.py
FSPL, EIRP, margin
Ka/X/S/UHF bands
CCSDS 141.0-B-1

ROUTING

rl_agent.py
bundle.py / ltp.py
Q-learning, federated
BPv7, LTP, TCPCL, UDP-CL
RFC 9171, RFC 5326

SECURITY

qkd.py
repeater_chain.py
BB84, E91 protocols
CASCADE reconciliation
NIST FIPS 203/204 (PQC)

ORBITAL

contact_windows.py
topology.py / doppler.py
5-tier, 241 nodes
Synodic period modeling
Doppler compensation

SIMULATION ENGINE

simulator.py · policy_engine.py · training.py · multi_agent.py · forwarding_engine.py

WEB SHOWCASE – 10 Interactive Demos

matx104.github.io/AETHERIX · 149 tests · 27 Python modules

CCSDS 734.2-B-1 · CCSDS 735.1-B-1 · CCSDS 141.0-B-1

RFC 9171 · RFC 5326 · RFC 7242 · NIST FIPS 203/204

BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

APPLICATION LAYER

Science data, commands, imagery

BUNDLE PROTOCOL v7 (RFC 9171)

Store-and-Forward | Custody Transfer | Priority (P0–P4)

CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

PHYSICAL LAYER

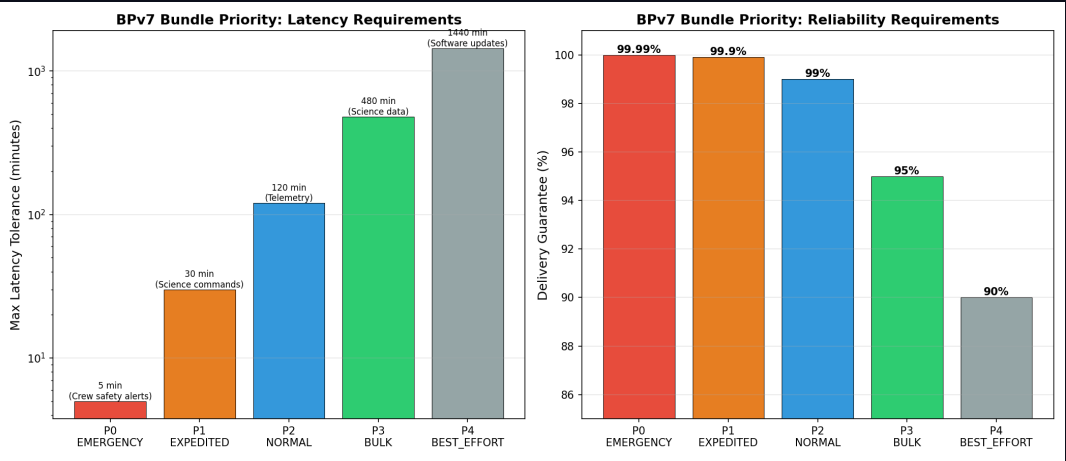
Optical 1550nm (2–200 Mbps) | RF Ka-band | UHF (surface)

HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

STANDARDS COMPLIANCE

- RFC 9171 — Bundle Protocol Version 7
- RFC 4838 — DTN Architecture
- RFC 5326 — Licklider Transmission Protocol
- RFC 9172 — Bundle Protocol Security (BPsec)
- CCSDS 734.2-B-1 — CCSDS Bundle Protocol
- CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing



DTN STORE-AND-FORWARD

How Bundles Traverse the Interplanetary Network

TCP/IP FAILS

- Timeout at 44 min RTT → connection reset
- No persistent path → routing impossible
- ACKs take minutes → window collapse
- High BER → retransmission storms

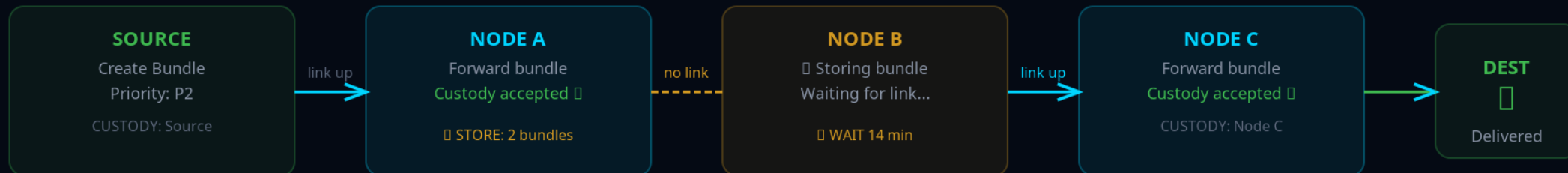
DTN WORKS

- No end-to-end connection required
- Store locally → wait for contact window
- Custody transfer: hop-by-hop reliability
- Priority classes: P0–P4 for urgency

TCP/IP: END-TO-END CONNECTION (FAILS IN SPACE)



DTN: STORE-AND-FORWARD (WORKS IN SPACE)



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Three Convergence Layers

LTP (RFC 5326) — Deep space

TCPCL (RFC 7242) — Earth segment

UDP-CL — Optical ISL

5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

TIER 3: Deep Space (4 nodes)

ES-L4, ES-L5 Lagrange relays + Transfer orbit

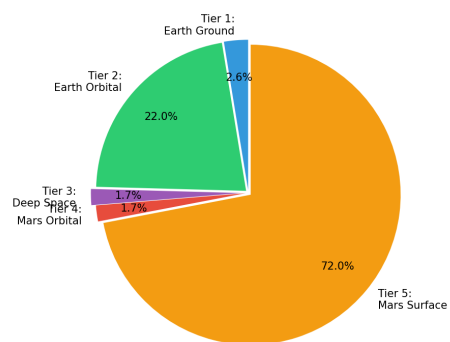
TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

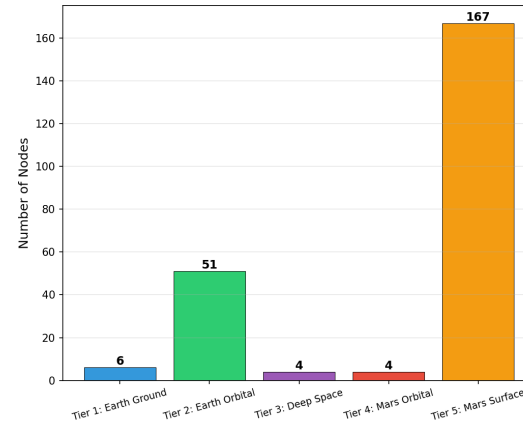
TIER 5: Mars Surface (176 nodes)

Bases, rovers, drones, sensor network

AETHERIX Network Node Distribution



Total Network: 232 Nodes



relays provide coverage
to-end key distribution.

5 · MARS SURFACE TIER 4 · MARS ORBITAL TIER 3 · DEEP SPACE TIER 2 · EARTH ORBITAL TIER 1 · EARTH GROUND



PROTOCOLS

BPv7	RFC 9171
LTP	CCSDS 734.2
QKD	RL Routing

LEGEND

- Earth Tiers
- Deep Space Transit
- Mars Orbital
- Mars Surface
- Cross-link
- Inter-tier

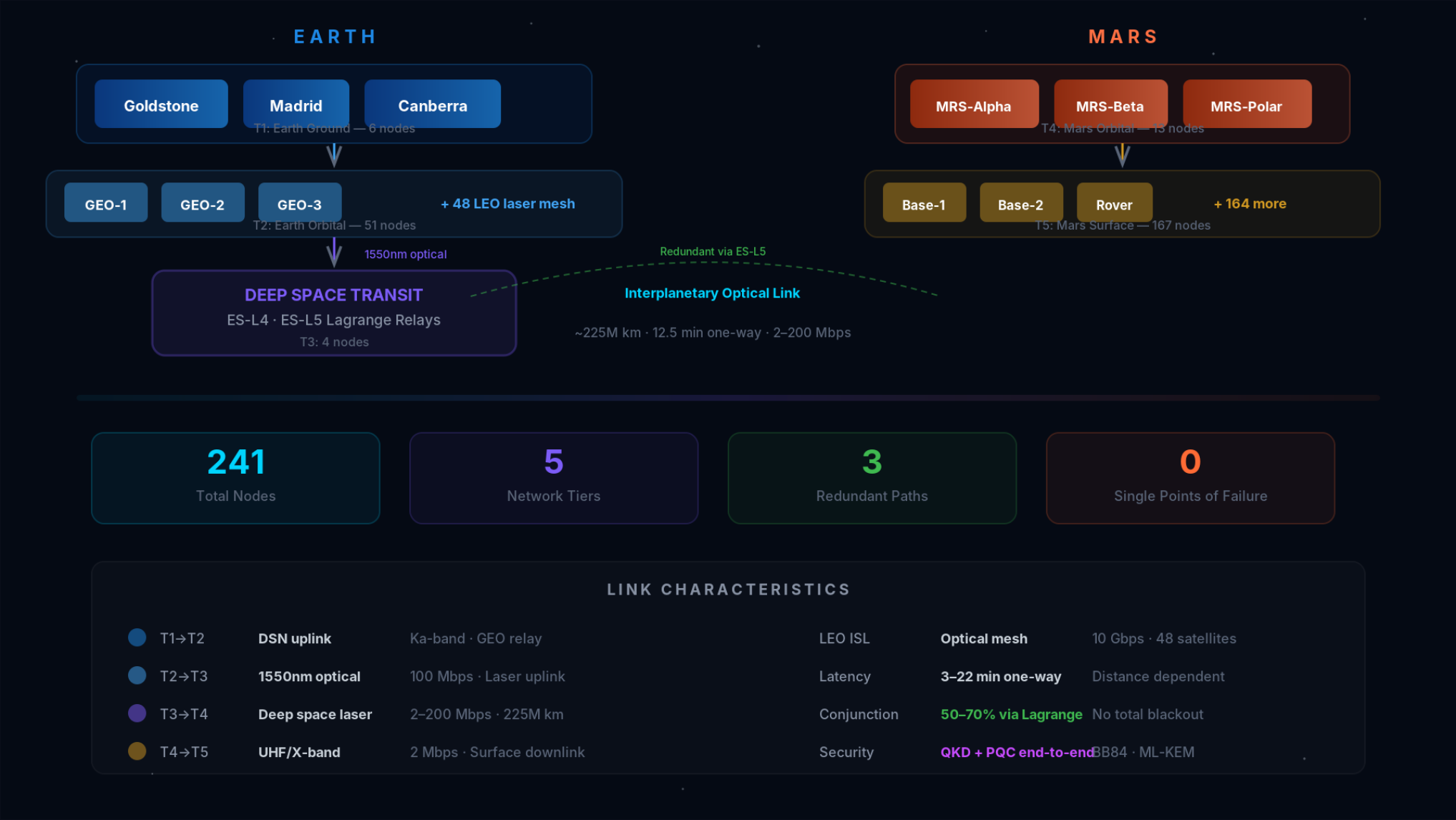
EARTH–MARS DISTANCE

54.6 M km (perihelion)
401 M km (aphelion)

AETHERIX · Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange · CCSDS / RFC Compliant

5-TIER NETWORK DIAGRAM

241 Nodes from Earth to Mars Surface



OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Faster Than RF

200 Mbps

Peak Data Rate

At perihelion (54.6M km)

1550 nm

Wavelength

Telecom heritage lasers

33×

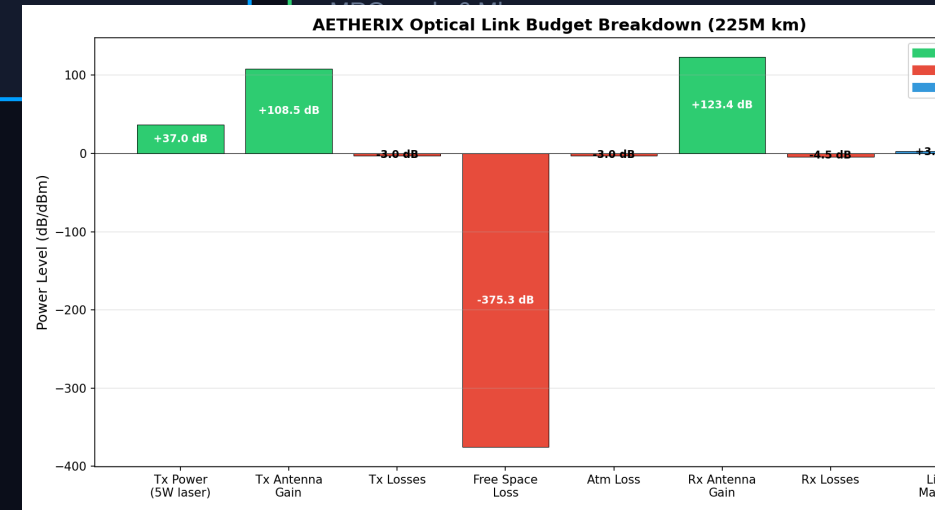
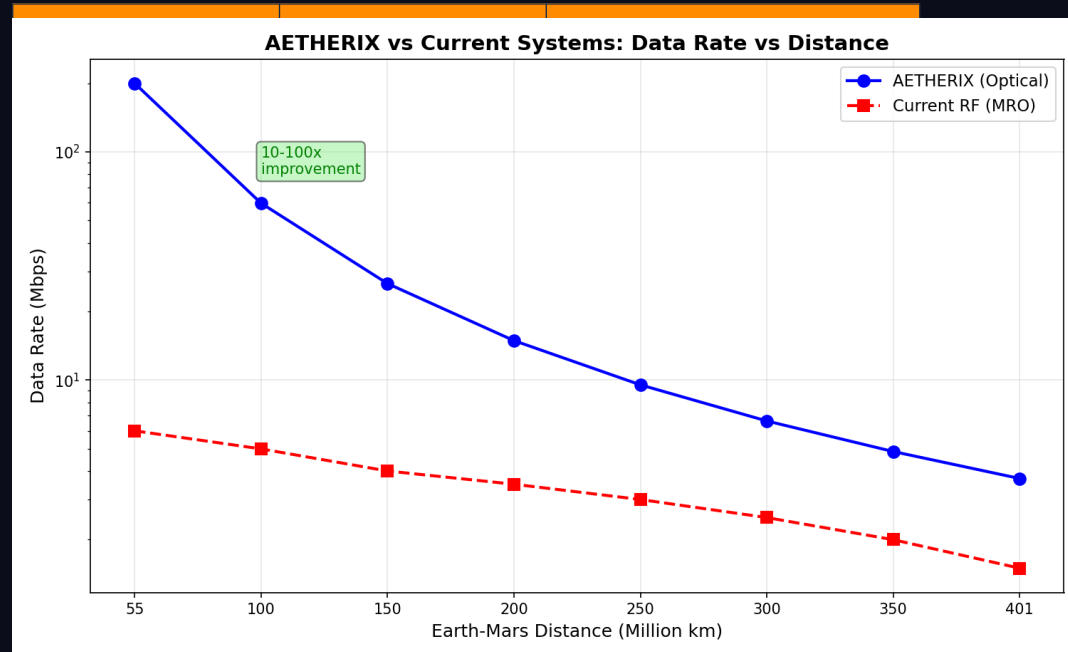
vs Current RF

KEY EQUATIONS

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss

$G = (\pi D/\lambda)^2$ Antenna Gain

$P_r = P_t \cdot G_t \cdot G_r / FSPL$ Received Power



EARTH-MARS DATA JOURNEY

7 Hops from Perseverance Rover to Mission Control

Hop	From	To	Link	Time
1	Perseverance	MRS-Alpha	UHF	~5s
2	MRS-Alpha	MRS-Polar	Optical ISL	~1s



HOP DETAIL



Total Transit: ~13 min
vs 12.5 min light-time!

DTN Overhead: <5%
Store-and-forward adds minimal latency

QKD Secured
End-to-end quantum encryption

☐ If any link drops mid-transfer, the bundle is stored safely — zero data loss

RL-BASED ROUTING — AI INNOVATION

Multi-Agent Federated Q-Learning Replaces Static CGR

CGR LIMITATIONS

- Static contact plan — no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

RL AGENT ADVANTAGES

- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- Reward: $\alpha(\text{delivery}) - \beta(\text{delay}) - \gamma(\text{hops})$
- Federated learning across agents
- Auto-recovery in seconds, not hours

97%

Delivery Ratio

+5% vs CGR

-20%

Delivery Time

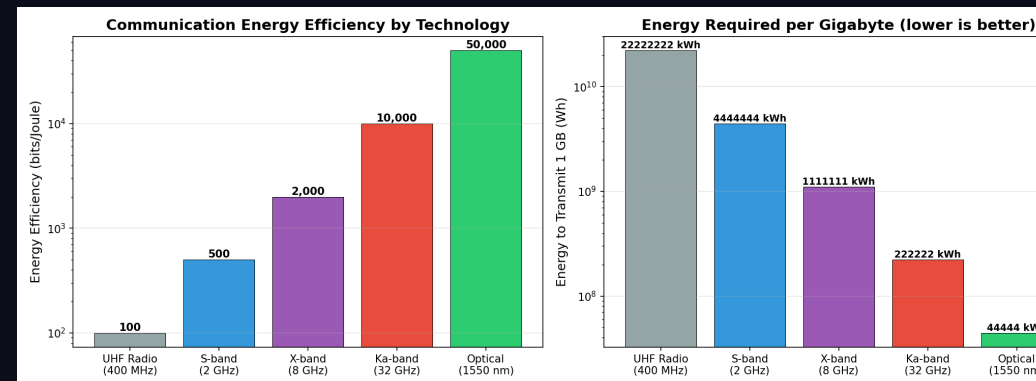
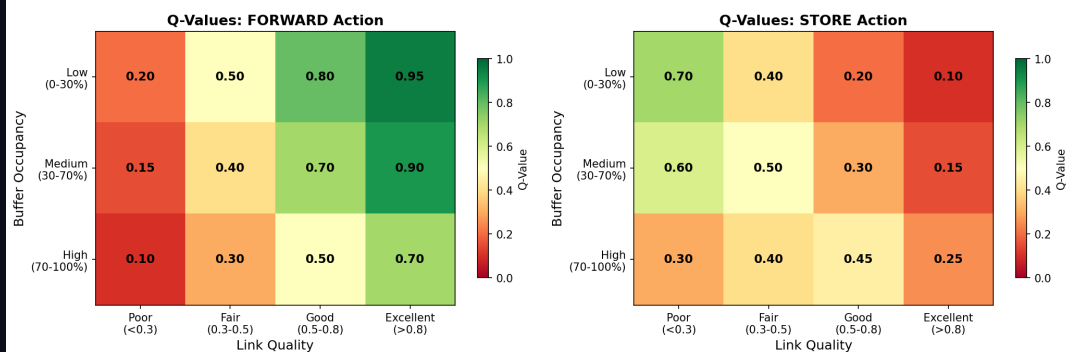
1.2× light-time

3600×

Recovery Speed

Auto vs manual

AETHERIX RL Routing Agent: Decision Heatmaps



QUANTUM SECURITY — FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

BB84 PROTOCOL STEPS

1. Alice sends qubits in random bases (rectilinear/diagonal)
2. Bob measures each qubit in random basis
3. Public reconciliation of basis choices (~50% retained)
4. QBER estimation: $< 11\%$ = SECURE, $\geq 11\%$ = ABORT
5. CASCADE error reconciliation
6. Privacy amplification via universal hashing
7. Final shared secret key

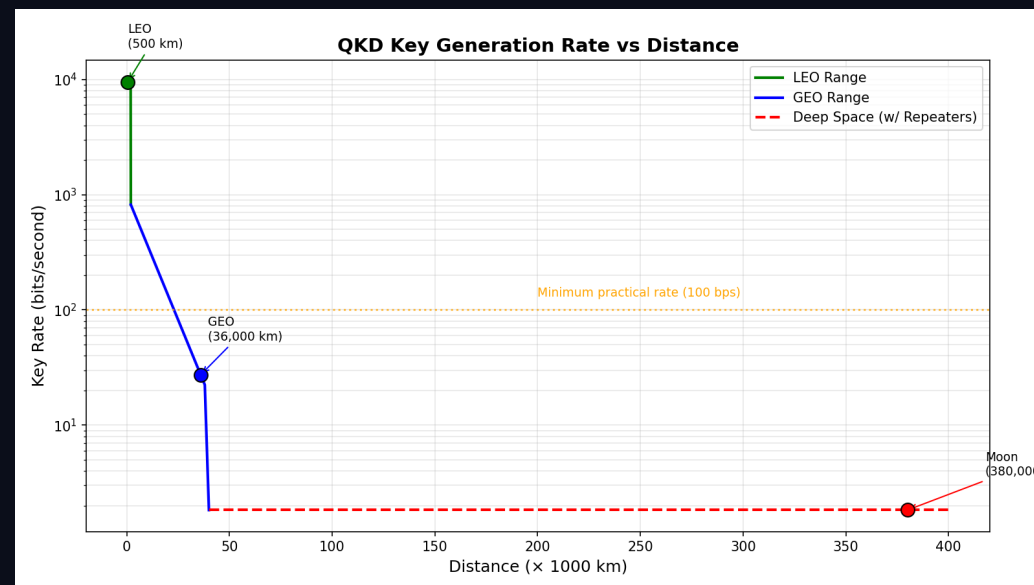
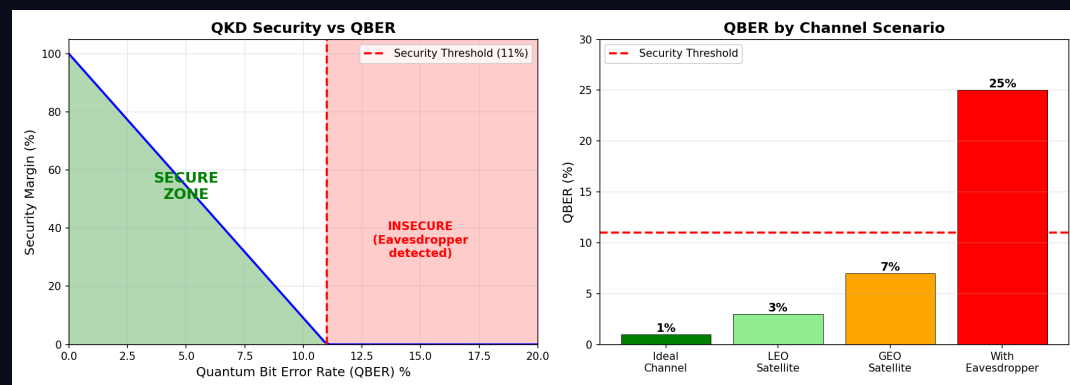
Phase	Segment	Protocol	Range
Phase 1	Earth $\leftrightarrow$ Orbit	BB84 direct	36,000 km
Phase 2	Earth $\leftrightarrow$ Lagrange	1 repeater	1.5M km
Phase 3	Earth $\leftrightarrow$ Mars	3-hop chain	225M km

POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM) — Key encapsulation

Dilithium (ML-DSA) — Digital signatures

Hybrid: QKD + PQC for defense in depth

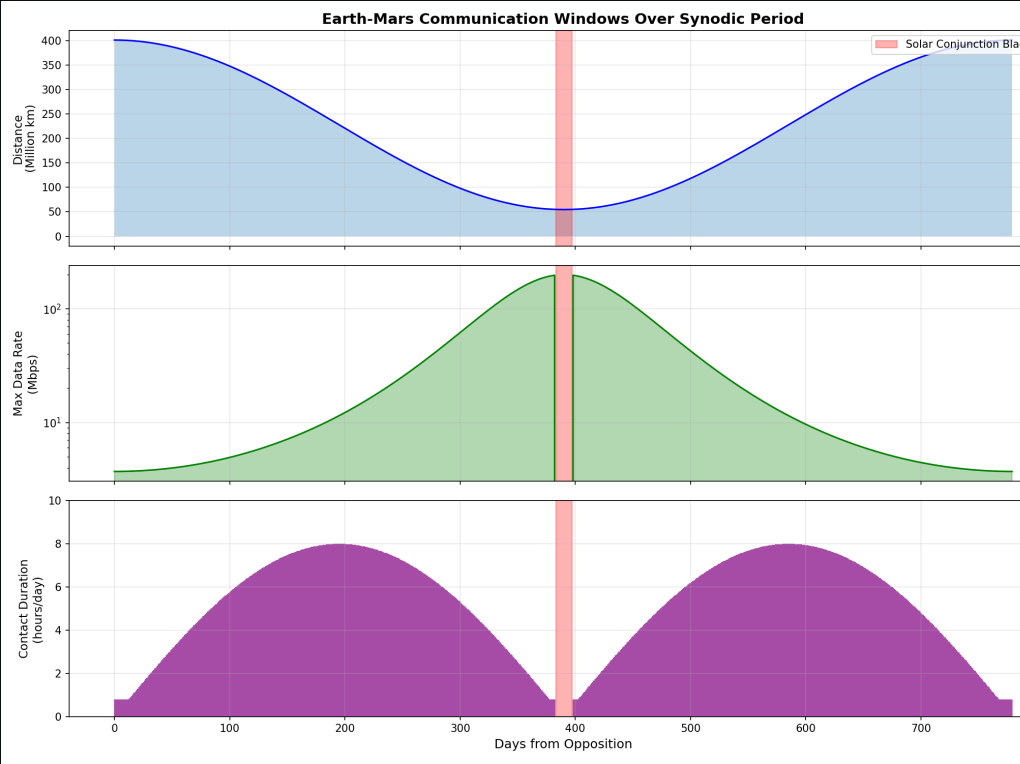
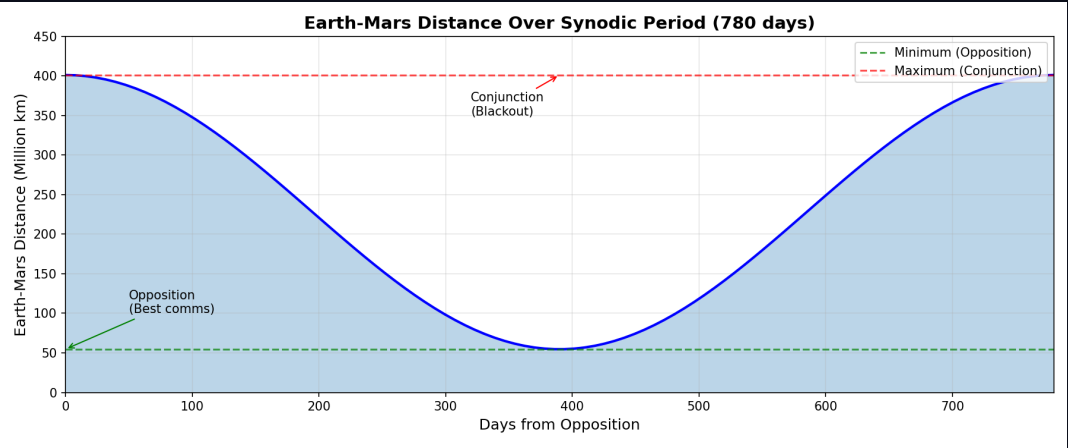


ORBITAL MECHANICS & CONTACT WINDOWS

780-Day Synodic Period — From Opposition to Conjunction

Parameter	Value
Semi-major axis	1.524 AU (227.9M km)
Orbital period	686.98 days
Synodic period	779.94 days
Min distance	54.6M km (perihelion)
Max distance	401M km (aphelion)
Eccentricity	0.0934

Window	Availability	Duration	Data Rate
Optimal (opposition)	99%	8–12 hrs	100–200 Mbps
Good	95%	6–8 hrs	20–100 Mbps
Fair	85%	2–4 hrs	5–20 Mbps
Poor (aphelion)	50%	0–2 hrs	1–5 Mbps
Blackout (conjunction)	0%	N/A	Lagrange relay only



RADIATION-HARDENED COMPUTING

Surviving SEUs, latchup and total dose en route to Mars

Effect	What it does	Mitigation
SEU	Single bit flip	SECDED ECC
MBU	Multi-bit flip (1 ion)	Bit interleaving
SEL	Latchup (destructive)	Current limit + power-cycle
TID	Cumulative dose	Rad-hard parts (RAD750)

DEFENSE-IN-DEPTH STACK

- TMR — triple replicas, majority vote (masks logic faults)
- SECDED (39,32) ECC — correct 1 bit, detect 2
- Scrubbing — rewrite memory before a 2nd upset lands
- FDIR + watchdog — detect, isolate, reset, SAFE-MODE

200x

Fewer errors (ECC + scrub + interleave)

3,334x

TMR reliability gain ($p=1e-4/op$)

200 krad

RAD750 TID tolerance (>2000x margin)

~0.9/day

Residual uncorrectable, Mars transit

Model: 512 Mbit, ~210-day GCR cruise. ~37,000 raw bit upsets reduced to ~186 uncorrectable over the mission. Heritage: NASA RAD750 (Curiosity/Perseverance), ESA LEON3FT. -> [src/computing/radiation.py](#)

MISSION-CRITICAL DATA PRIORITIZATION

Bandwidth triage: get the right bits home first

Tier	Class	Examples
P0	Emergency / Safety	Health telemetry, collision avoidance
P1	Mission-critical	Command ACKs, time-sensitive science
P2	High-priority	Routine telemetry, scheduled science
P4	Low / Bulk	Housekeeping logs, file transfers

Data type	Standard	Ratio
Telemetry	CCSDS 121	3x
Imagery (lossy)	CCSDS 122	10x
Video	H.265	50x

100%

Link utilization (no wasted bandwidth)

5 / 6

Items fully delivered by priority

BPv7

Fragmentation defers bulk remainder

Preempt

Emergency uses direct-to-Earth backup

Scenario: 30 Mbps, 15-min contact, oversubscribed. Deadline-aware, preemptive QoS scheduler delivers emergency + mission + science first; 6 GB software update fragmented to the next pass. -> src/routing/prioritization.py

END-TO-END MISSION SCENARIO

Perseverance 500 MB Science Data Upload

Hop	Node	Action	Protocol	Time
1	Perseverance	Create bundle	BPv7	T+0s
2	MRS-Alpha	Store & forward	UHF	T+5s
3	MRS-Polar	PL route select	Optical ISL	T+10s



HOP DETAIL



Total Transit: ~13 min
vs 12.5 min light-time!

DTN Overhead: <5%
Store-and-forward adds minimal latency

QKD Secured
End-to-end quantum encryption

✓ **If any link drops mid-transfer, the bundle is stored safely — zero data loss**

DATA FLOW — APPLICATION TO DELIVERY

How Data Travels Through the AETHERIX Stack

APPLICATION → TRANSPORT

1. CREATE

Application generates payload (science data, commands)

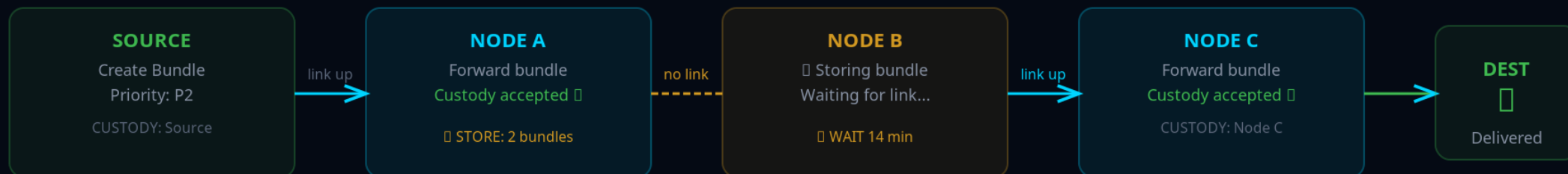
2. ENCODE

BPv7 primary block + payload block + security block

TCP/IP: END-TO-END CONNECTION (FAILS IN SPACE)



DTN: STORE-AND-FORWARD (WORKS IN SPACE)



Key: Custody Transfer

Each node takes responsibility for the bundle

Data Never Lost

Bundle stored until next hop confirms

No End-to-End Needed

Hop-by-hop, works with 22-min delays

Three Convergence Layers

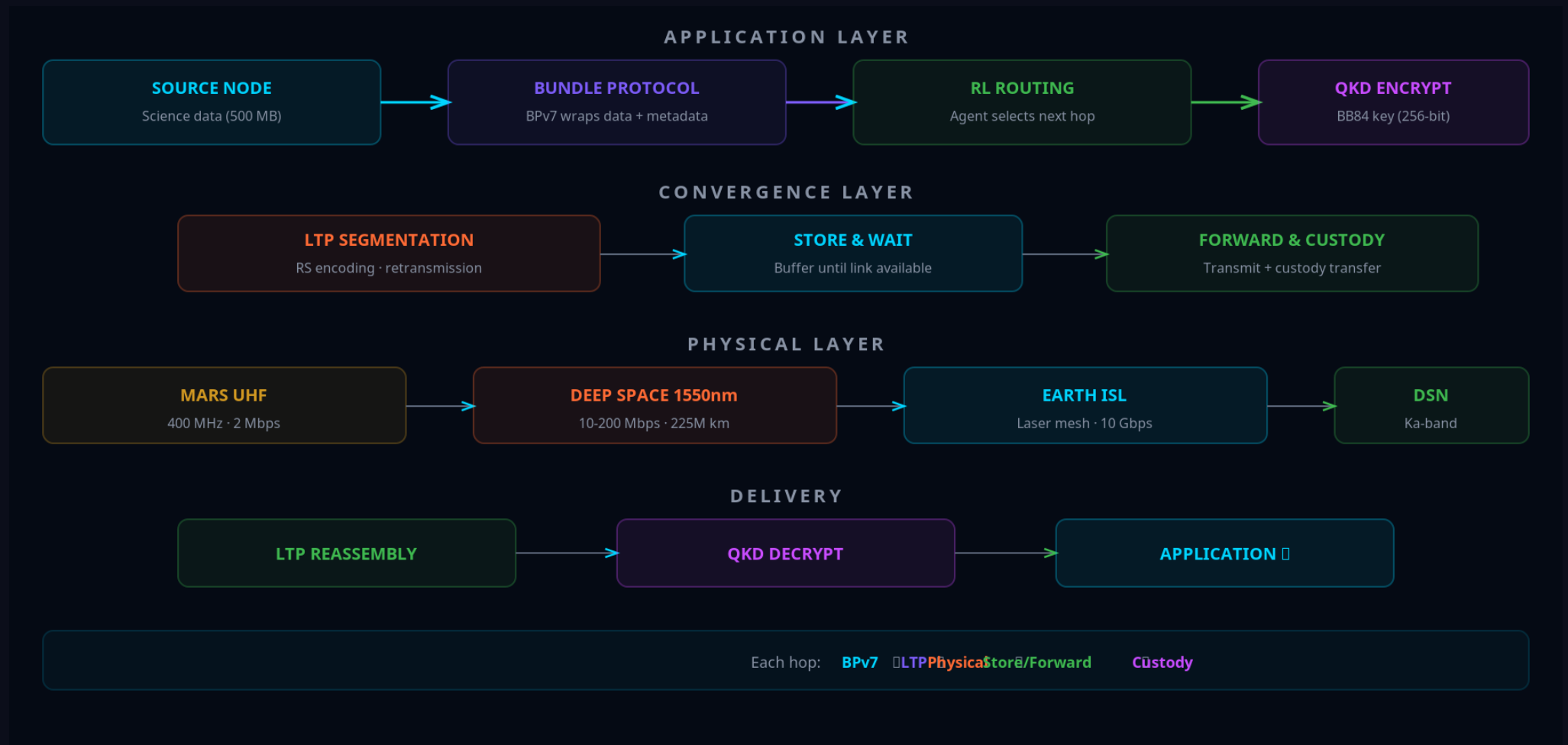
LTP (RFC 5326) — Deep space

TCPCL (RFC 7242) — Earth segment

UDP-CL — Optical ISL

DATA FLOW DIAGRAM

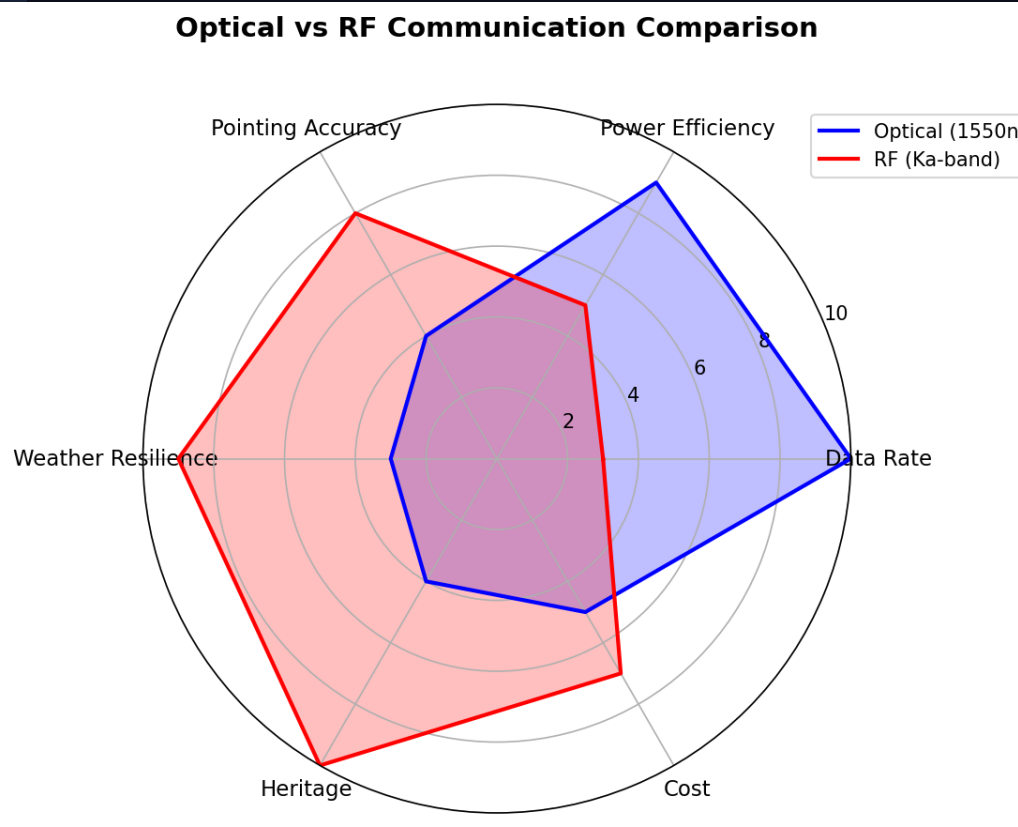
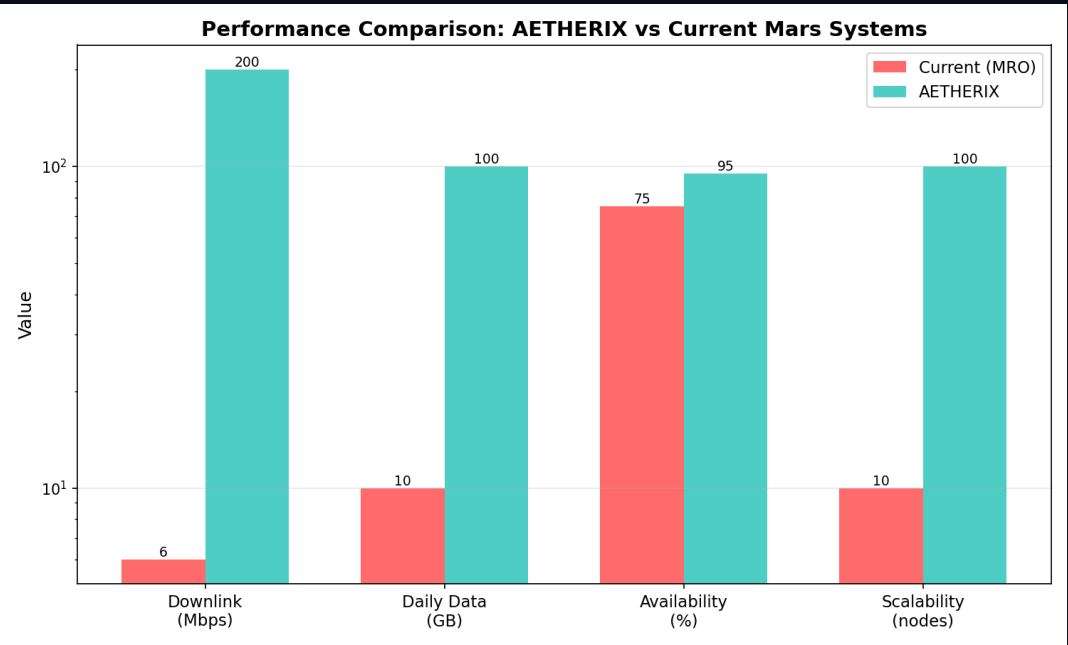
Complete Data Path from Mars Surface to Earth Control



AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Comparison

Metric	Current (MRO)	AETHERIX	Improvement
Downlink rate	0.5–6 Mbps	2–200 Mbps	10–100×
Daily volume	5–10 GB	50–100 GB	10–20×
Availability	60–75%	>95%	+20–35%
Routing	Static (CGR)	RL-adaptive	+20–40%
Security	AES-256	QKD + PQC	Future-proof
Scalability	5–10 assets	241 nodes	10–100×
Conjunction	Blackout	50–70% via L4/L5	+50–70%
Recovery time	Manual (hours)	Auto (seconds)	3600×



IMPLEMENTATION

18 Modules, 149 Tests, Full Python Stack

18

Source Modules

149

Unit Tests

241

Network Nodes

6

Interactive Demos

CORE MODULES

- link_budget.py — Optical/RF link analysis
- rl_agent.py — Q-learning routing agent
- bundle.py — BPv7 bundle protocol
- forwarding_engine.py — Store-and-forward
- qkd.py — BB84/E91 protocols
- contact_windows.py — Orbital mechanics
- topology.py — 5-tier network (241 nodes)
- simulator.py — End-to-end simulation
- ltp.py — Licklider Transmission Protocol
- tcpcl.py — TCP Convergence Layer

Standard	Title	Status
RFC 9171	Bundle Protocol v7	Compliant
RFC 4838	DTN Architecture	Compliant
CCSDS 734.2-B-1	CCSDS Bundle Protocol	Compliant
CCSDS 734.3-B-1	SABR / contact routing	Baseline
CCSDS 141.0-B-1	Optical Comm Physical	Compliant
RFC 9172	BPsec (security)	Referenced
RFC 5326	LTP (deep space)	Compliant
RFC 7242	TCPCL (Earth)	Compliant

ROADMAP & FUTURE WORK

From Proof-of-Concept to Production Deployment

Phase 1–2: Foundation

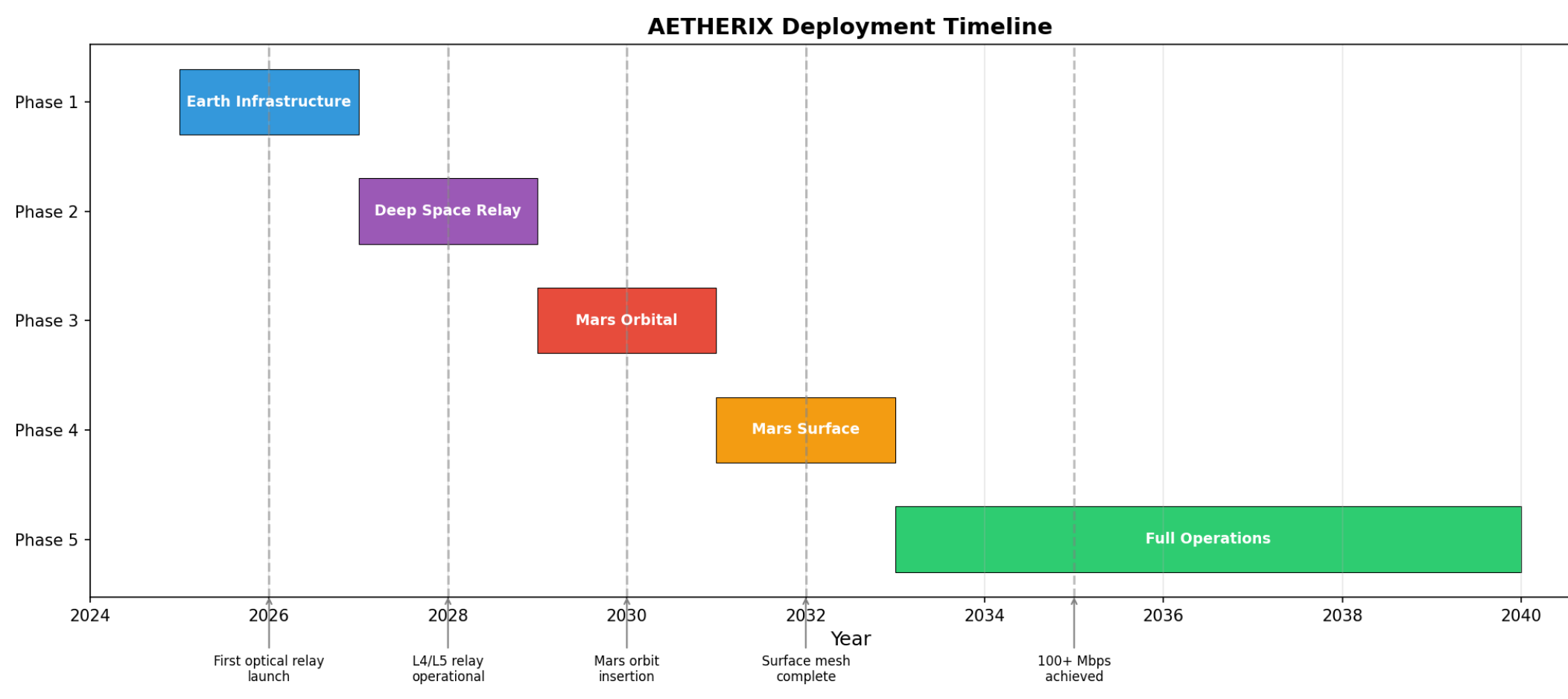
Topology + BPv7 + Convergence Layers

Complete

Phase 3: Intelligence

Complete

Phase	Focus	Technology	Status
7	DQN Upgrade	Deep Q-Network	Remaining
8	HIL Simulation	ns-3 integration	Remaining
9	ION-DTN	Production DTN stack	Remaining
10	Flight Demo	LEO optical test	Future
11	Mars Deploy	Full 5-tier network	Future



CONCLUSION — AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

THE PROBLEM

- 54.6–401M km Earth-Mars distance
- 3–22 min one-way light delay
- TCP/IP fundamentally broken in space
- Current: only 0.5–6 Mbps
- 2-week solar conjunction blackouts

THE SOLUTION

- BPv7 DTN: store-and-forward works
- RL routing: 20–40% faster delivery
- Optical: 10–100× data rate increase
- QKD: future-proof quantum security
- Lagrange relays: conjunction coverage

10–100×

Faster Rates

>95%

Availability

AI-Driven

Routing

Quantum

Security

NOVEL CONTRIBUTIONS

RL autonomous routing (20-40% faster) | 5-tier DTN (>95% availability) | Optical/RF hybrid (10-100×) | Quantum links (future-proof) | Full simulation (149 tests)

THANK YOU

Questions?

10–100×

Faster

>95%

Available

241

Nodes

149

Tests

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LIVE DEMO & RESOURCES

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

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